

ANSWERS MIDTERM FIRST PART

Question:

- (1) State the Cauchy-Schwartz Inequality and the Triangle Inequality (and the equivalent conditions for equality in both cases).
- (2) Verify the Cauchy Schwartz Inequality and the Triangle Inequality for the vectors $v = (3, -1, 2)^T$ and $w = (1, -1, 1)^T$ using the inner product $\langle v, w \rangle = v^T K w$ where K is the matrix

$$K = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

Answer:

The *Cauchy-Schwartz inequality* states that, if V is a real vector space with an inner product $\langle, \rangle$ and associated norm $\|\cdot\|$, then for any two vectors $v, w \in V$, $|\langle v, w \rangle| \leq \|v\| \|w\|$, with equality if and only if $v = \lambda w$ for some $\lambda \in \mathbb{R}$ or $w = \mu v$ for some $\mu \in \mathbb{R}$.

The *Triangle Inequality* states that, if V is a real vector space with an inner product $\langle, \rangle$ and associated norm $\|\cdot\|$, then for any two vectors $v, w \in V$, the associated norm satisfies $\|v + w\| \leq \|v\| + \|w\|$, with equality if and only if $v = \lambda w$ for some real number $\lambda \geq 0$ or $w = \mu v$ for some real number $\mu \geq 0$.

In the above case,

$$\langle v, w \rangle = 2v_1w_1 - v_1w_2 - v_2w_1 + 2v_2w_2 - v_2w_3 - v_3w_2 + 2v_3w_3$$

and

$$\|v\|^2 = 2v_1^2 - 2v_1v_2 + 2v_2^2 - 2v_2v_3 + 2v_3^2$$

So $\langle v, w \rangle = 6 + 3 + 1 + 2 + 2 + 1 + 4 = 19$ and $\|v\|^2 = 18 + 6 + 2 + 4 + 8 = 38$ and $\|w\|^2 = 2 + 2 + 2 + 2 + 2 = 10$. Hence $|\langle v, w \rangle| = 19 \leq \|v\| \|w\| = \sqrt{38} \sqrt{10} = \sqrt{380}$. This follows from squaring both sides and noting that $361 \leq 380$. This verifies the Cauchy-Schwartz Inequality.

We have $v + w = (4, -2, 3)$, hence $\|v + w\|^2 = 32 + 16 + 8 + 12 + 18 = 86$, hence $\|v + w\| = \sqrt{86} \leq \|v\| + \|w\| = \sqrt{38} + \sqrt{10}$. This follows from squaring both sides and noting that $86 = 38 + 10 + (2 \times 19) < 38 + 10 + (2 \times \sqrt{380})$. This verifies the Triangle Inequality.

Question:

- (1) Define the notion of an open set, a closed set, a bounded set and a compact set in $\mathbb{R}^n$.
- (2) State the Heine-Borel Theorem.

- (3) Consider the set $A = \{(x, y) \in \mathbb{R}^2 \mid 0 < x^2 + y^2 \leq 1\}$. Is A a closed subset of $\mathbb{R}^2$? Is A an open subset of $\mathbb{R}^2$? Briefly justify your answers.

Answer:

A set $A \subset \mathbb{R}^n$ is said to be *open* if for every $a \in A$ there exists some $r > 0$ (depending on a) satisfying $B(a; r) \subset A$. Here, given $a \in \mathbb{R}^n$ and $r > 0$, $B(a; r)$ is defined as $B(a; r) = \{x \in \mathbb{R}^n \mid \|x - a\| < r\}$.

A subset $A \subset \mathbb{R}^n$ is *closed* if A contains all its limit points, where a *limit point* of A is a point $x \in \mathbb{R}^n$ such that there exists a convergent sequence $\{a_n\}$ of points in A whose limit is x , that is $\lim_{n \rightarrow \infty} a_n = x$.

A set $A \subset \mathbb{R}^n$ is *bounded* if there exists $R > 0$ (depending on A) such that $A \subset B(0; R)$.

A set $A \subset \mathbb{R}^n$ is *compact* if any sequence $\{a_n\}$ in A has a convergent subsequence $\{a_{n_k}\}$ whose limit belongs to A , that is $\lim_{k \rightarrow \infty} a_{n_k} \in A$.

The *Heine-Borel Theorem* states that a subset A of $\mathbb{R}^n$ is compact if and only if it is a closed and bounded subset of $\mathbb{R}^n$.

The given set of $A \subset \mathbb{R}^2$ is neither an open subset of $\mathbb{R}^2$ (since no neighbourhood of any point of the form (x, y) with $x^2 + y^2 = 1$, belongs to A) nor a closed subset of $\mathbb{R}^2$ (since $(0, 0)$ is a limit point of A which does not belong to A).

ANSWERS MIDTERM SECOND PART

Question: What is a positive definite matrix? What is the Cholesky factorization of such a matrix? Find the Cholesky factorization of the following matrix

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Answer: A *positive definite matrix* A is a real n by n matrix (for some $n > 0$) which is a symmetric matrix ($A = A^t$) and such that $x^t A x > 0$ for all $x \in \mathbb{R}^n - \{(0, \dots, 0)\}$.

The *Cholesky factorization of a positive definite matrix* A is the decomposition $A = MM^t$ where M is a lower triangular matrix with positive entries on the diagonal (hence M^t is an upper triangular matrix with positive entries on the diagonal). The matrix M (and hence M^t) is uniquely determined for a given positive definite matrix A .

The lower triangular matrix of the Cholesky decomposition of the given matrix is

$$\begin{bmatrix} \sqrt{2} & 0 & 0 \\ \frac{1}{\sqrt{2}} & \frac{\sqrt{3}}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{3}} \end{bmatrix}$$

Question: What is an orthogonal matrix? What is the QR decomposition of a nonsingular matrix? Find the QR decomposition of the following matrix

$$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 5 & 3 \\ 3 & 0 & -1 \end{bmatrix}.$$

Answer: An *orthogonal matrix* A is a real n by n matrix (for some $n > 0$) such that $A^t A = Id_n = A A^t$.

The *QR decomposition of a nonsingular real n by n matrix* A is the decomposition $A = QR$ where Q is an orthogonal n by n matrix and R is an upper triangular real n by n matrix with positive entries on the diagonal. The matrices Q and R are uniquely determined for a given nonsingular matrix A .

For the matrix A above, we have

$$Q = \begin{bmatrix} \frac{1}{\sqrt{14}} & \frac{1}{\sqrt{3}} & -\frac{5}{\sqrt{42}} \\ \frac{2}{\sqrt{14}} & \frac{1}{\sqrt{3}} & \frac{4}{\sqrt{42}} \\ \frac{3}{\sqrt{14}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{42}} \end{bmatrix}$$

and

$$R = \begin{bmatrix} \sqrt{14} & \sqrt{14} & \frac{5}{\sqrt{14}} \\ 0 & 3\sqrt{3} & 2\sqrt{3} \\ 0 & 0 & \frac{\sqrt{3}}{\sqrt{14}} \end{bmatrix}$$